

Harris County Flood Control District

Hydrology and Hydraulics Design Manual

Section 2.3: Detention Sizing for Watershed Developments

Development within Harris County watersheds requires strict adherence to localized flood control standards. For commercial developments located within the Brays Bayou watershed, detention facilities must be sized to manage the 100-year storm event. To account for local soil saturation characteristics, volume sizing must utilise the Tabular Unit Hydrograph method, applying the area, unit runoff, and impervious factor parameters listed in Table 2.3a.

$$\text{Required Volume (cu ft)} = \text{Area} \times \text{Unit Runoff} \times \text{Impervious Factor}$$

where Area is the site area in acres, Unit Runoff is read from Table 2.3a for the applicable storm event and soil hydrologic group, and the Impervious Factor is read from the impervious coverage schedule.

Table 2.3a: Brays Bayou Watershed Design Parameters

Parameter	Value
100-Year Storm Unit Runoff, Type D soils	13,600 cu ft/acre
Impervious Factor, 80% coverage	1.25

These parameters apply specifically to sites within the Brays Bayou watershed. The Type D hydrologic soil group reflects the clay-dominated soils characteristic of this watershed, which exhibit low infiltration and high runoff potential.